Predicting Risky Choices for New Participants:
CS-3410 / MAT-3211-1 Midterm Project, Track B (Behavioural Economics)

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Abstract

All data in this project are *simulated*; conclusions concern the simulated system and the methods used, not real people. I predict whether a simulated participant chooses a risky prospect over a sure amount, and ask whether framing and prospect-theory-based features improve prediction for *unseen* participants. With participant-grouped cross-validation, logistic regression on the four raw numbers reaches ROC-AUC 0.721, the expected-value rule 0.727, adding expected-value features, domain and frame 0.812, and adding prospect-theory features 0.817 (fold standard deviation 0.018, so the last step is within noise). A random split is not optimistic for this model (0.818 vs. 0.817) unless the model may use the participant identifier (0.840 vs. 0.817).

1 Research question

Question. For *new* participants, do (i) the frame and domain of a choice problem and (ii) features motivated by prospect theory improve the ranking of risky choices over a model that sees only the four problem numbers (probability, two outcomes, sure amount)?

Why it matters. The hidden test set contains 90 participants never seen in training, so a model, and especially a validation scheme, that implicitly learns individuals will look better than it is. The question tests both which features carry transferable signal and how optimistic naive validation is.

Hypotheses

H1. Expected-value (EV) features, domain and frame improve AUC; person details (age, numeracy, risk score) do not help for unseen people.

H2. A random row-level split gives a higher (too optimistic) AUC than a participant split, mainly if the model can recognise a person.

H3. The response time, recorded together with the choice, adds nothing; I exclude it anyway because it would not be known before choosing.

H4. Prospect-theory (PT) value-difference features add a little beyond EV features, because the data come from a stochastic PT model.

2 Background and related work

Expected-value maximisation predicts that people take the option with the higher average payoff, yet people depart from this systematically. Tversky and Kahneman's cumulative prospect theory [1] describes choice with a value function that is concave for gains, convex for losses and steeper for losses (loss aversion, factor λ), and a probability-weighting function that overweights small probabilities. The same final-wealth problem is chosen differently when presented as gains or as losses [2]; the frame, invisible to an EV rule, should therefore matter for prediction. I use this literature only to design *fixed* features (Section 4); no parameter is fitted to the training labels.

On the methods side, observations from the same participant are correlated, so a random split places the same person in training and validation. Grouped cross-validation avoids this [3]. Any

column that would not exist at prediction time is leakage [4]; I identify and drop such columns in Section 3. Because the primary metric ranks probabilities, I also check calibration of the probabilities [5]. All models are standard scikit-learn implementations [6].

3 Data and preprocessing

The training file has 8,400 decisions of 210 participants (40 each); the test file has 3,600 decisions of 90 participants. No participant occurs in both files. The target `chose_risky` is balanced (48.2% risky), so ROC-AUC (the competition metric) is appropriate. Risky-choice rates depend strongly on domain and frame (Figure 1, left): 39.8% for gains in the gain frame, 55.7% for the *same* gain problems in the loss frame, 58.7% for losses and 38.3% for mixed gambles.

Leakage audit. `realised_payout_inr` exists only in the training file (4,600 missing values) and is determined after the choice; it is dropped. `participant_id` is used only to define validation groups, never as a feature in the final model. `response_time_ms` is excluded from the final model (it is measured with the choice; Experiment 3 checks that this loses nothing). Identifier columns are not features.

Preprocessing. Numeric columns are standardised and the categorical columns (`problem_domain`, `frame`, and their combination `cell`) are one-hot encoded. Both steps sit inside a scikit-learn pipeline and are fitted on training folds only.

4 Methods and experimental design

Validation. All reported AUC values use 12-fold `GroupKFold` on `participant_id` with pooled out-of-fold predictions, so every participant's 40 decisions stay together, matching the unseen-participant test. The random-split comparison uses 8-fold shuffled `KFold` (seed 0). To quantify stability I also compute AUC separately on each of five grouped folds. The Kaggle test file is used once, for the final prediction.

Features (all fixed formulas of the problem numbers; p = probability of the high outcome, H, L = high and low outcome, S = sure amount, E = endowment).

- *Nominal:* p, H, L, S.
- *EV features:* $\text{ev} = pH + (1-p)L - S$; spread $H - L$; $\text{ev}/(H-L)$; $(S-L)/(H-L)$; indicators $p \le 0.10$ and $p \ge 0.90$ (overweighting of extreme probabilities).
- *Domain and frame:* one-hot domain, frame and their combination (cell), plus $\text{ev}/(H-L)$ separately for each of the four cells.
- *Prospect-theory features (new):* with $v(x) = x^a$ for $x \ge 0$ and $v(x) = -\lambda(-x)^a$ for $x < 0$, and $w(q) = q^\gamma/(q^\gamma + (1-q)^\gamma)^{1/\gamma}$,

$$\text{pt} = \frac{w(p)\, v(H) + w(1-p)\, v(L) - v(S)}{E+1},$$

  computed for four fixed parameter sets (a, λ, γ): (0.88, 2.25, 0.61), (0.70, 1.5, 0.70), (0.88, 1.0, 0.80) and (0.60, 2.5, 0.60). Outcomes are used *as shown to the participant* (negative deductions in the loss frame), so the frame enters the feature. This simplifies the original model (one γ, weight applied to each outcome probability). Each pt feature is also split by the four cells, and $\text{ev}/(H-L)$ and $\text{ev}/(E+1)$ are multiplied by age, numeracy, self-reported risk taking, time pressure and a paid-session indicator ("trait interactions").

The final model has 45 numeric columns plus the one-hot columns. **Models.** Baselines (same protocol, nominal features only): (1) constant prior predictor, (2) the EV rule (score = ev, no training), (3) logistic regression, (4) LDA, (5) Gaussian Naive Bayes, (6) linear SVM. The experiments use L2-regularised logistic regression ($C = 1$ for Exp. 1a–1e, $C = 0.1$ for the final model; $C \in [0.03, 1]$ gave the same AUC to three decimals). Only methods taught in lectures are used.

Controlled experiments. Each changes one thing.

Table 1: Baselines, nominal features, 12-fold participant-grouped CV (pooled out-of-fold). The constant predictor scores below 0.5 only because its fold-specific constants are slightly anti-correlated with the held-out fold's base rate; its true AUC is 0.5.

Model	ROC-AUC	Log-loss	Accuracy
Constant (prior)	0.458	0.693	0.518
Expected-value rule	0.727	–	0.664
Logistic regression	0.721	0.619	0.662
LDA	0.716	0.620	0.657
Gaussian Naive Bayes	0.626	0.687	0.590
Linear SVM	0.718	–	–

Table 2: Experiments 1–3 (logistic regression; 12-fold participant CV unless stated).

Run	Change	ROC-AUC	Log-loss	Acc.
1a	nominal only	0.721	0.619	0.662
1b	+ EV features	0.783	0.559	0.704
1c	+ domain and frame	0.812	0.529	0.733
1d	+ person/context info	0.810	0.530	0.729
1e	+ cell interactions (original)	0.814	0.524	0.733
1f	**+ PT features, trait interactions (final)**	**0.817**	**0.520**	**0.735**
2a	final model, random 8-fold split	0.818	0.518	0.733
2b	final model, participant split	0.817	0.520	0.735
2c	+ `participant_id`, random split	0.840	0.493	0.754
2d	+ `participant_id`, participant split	0.817	0.519	0.735
3	final model + response time	0.817	0.520	0.734

- **Exp. 1** (features): nominal → +EV features → +domain/frame → +person/context → +cell interactions → +PT features and trait interactions (1a–1f).
- **Exp. 2** (validation): random vs. participant split, with and without `participant_id` as a one-hot feature.
- **Exp. 3** (availability): add `response_time_ms` to the final model.

Seeds: `numpy` seed 0; `KFold random_state=0`; the other components are deterministic.

5 Results

Baselines (Table 1). The EV rule (0.727) is as good as any model that only sees the four raw numbers. This is expected: $pH + (1-p)L - S$ is a product of columns that a linear model cannot form. Naive Bayes is clearly worst (0.626), consistent with its independence assumption failing for strongly dependent numeric features.

Experiment 1 (Table 2). EV features give the largest step (0.721 → 0.783) and domain/frame add another 0.029 (0.812), supporting H1: the frame is something an EV rule cannot see. Person and context variables do not help for unseen people (0.810, slightly below 0.812), also as in H1. Cell interactions add 0.002, and the PT features with trait interactions add a further 0.003 (0.817). The standard deviation across five grouped folds is 0.018 (fold AUCs $0.834, 0.803, 0.805, 0.802, 0.843$), so these last two steps are *within noise*; H4 is therefore only weakly supported.

Experiment 2. For the final model, random and participant splits give the same AUC (0.818 vs. 0.817), so H2 is *not* confirmed for a model that cannot identify people. When `participant_id` is added, the random split rises to 0.840 while the participant split stays at 0.817: knowing a person helps only if that person was seen in training. The test set has only new people, so the participant split is the honest estimate.

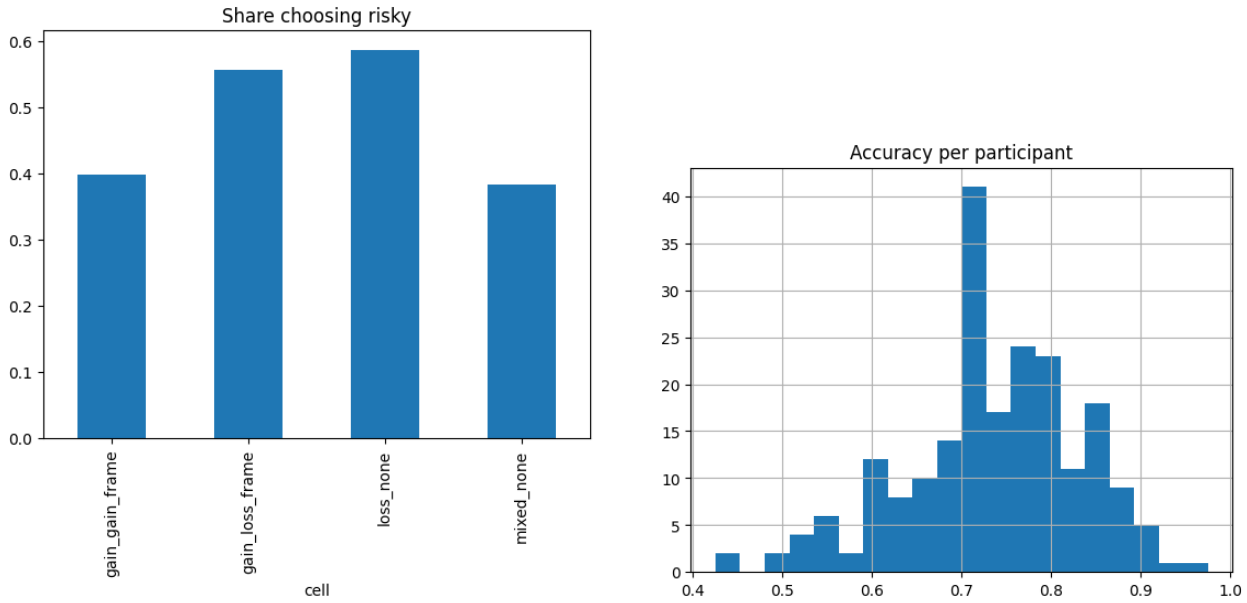


Figure 1: Left: share of risky choices by domain and frame (training data). The same gain problems are chosen risky more often in the loss frame. Right: out-of-fold AUC of the final model and the EV rule in each cell.

Experiment 3. Response time changes nothing (0.817), and its mean is almost equal for risky (823 ms) and safe (807 ms) choices; excluding it costs nothing.

Final model and leaderboard. The final model (Run 1f, $C = 0.1$) was fitted once on all training data and applied once to the test file, giving the prediction file `submission.csv` with probabilities (not 0/1 labels, since AUC uses the ranking). The cross-validated estimate 0.817 ± 0.018 is the expectation to compare with.

6 Error analysis and limitations

Where it works and where not (Figure 1, right). The final model beats the EV rule in every cell. The easiest cell is mixed gambles (AUC 0.898 vs. 0.858 for the EV rule), then losses (0.784 vs. 0.733), gains in the loss frame (0.765 vs. 0.722) and, hardest, gains in the gain frame (0.680 vs. 0.632). In the last cell choices are closest to a coin flip, so a limited achievable AUC is plausible, although I cannot verify this against the simulator.

Calibration (Figure 2, right). Predicted probabilities follow the diagonal closely, so they can be read as probabilities. AUC is higher in paid sessions (0.847) than in hypothetical ones (0.791); I do not claim a reason, because the simulator is unknown and this is an association.

People. Accuracy per participant ranges from 0.42 to 0.98 (mean 0.73). A model that never sees a new person's history cannot know whether they are a strong risk-taker, so the extremes are predicted worst. Person details do not fix this, as Experiment 1 shows.

Limitations.

- The data are simulated; no behavioural claim about real people follows.
- The simulator's choices are stochastic, so a perfect score is impossible; further feature engineering would give diminishing returns, and the improvements from 1d to 1f are within fold noise (0.018).
- Only 210 training participants; I use one CV run per model, and I selected the final model with the same CV, so 0.817 may be slightly optimistic.
- PT parameters are fixed, typical literature values (not the simulator's), and the PT form is simplified.

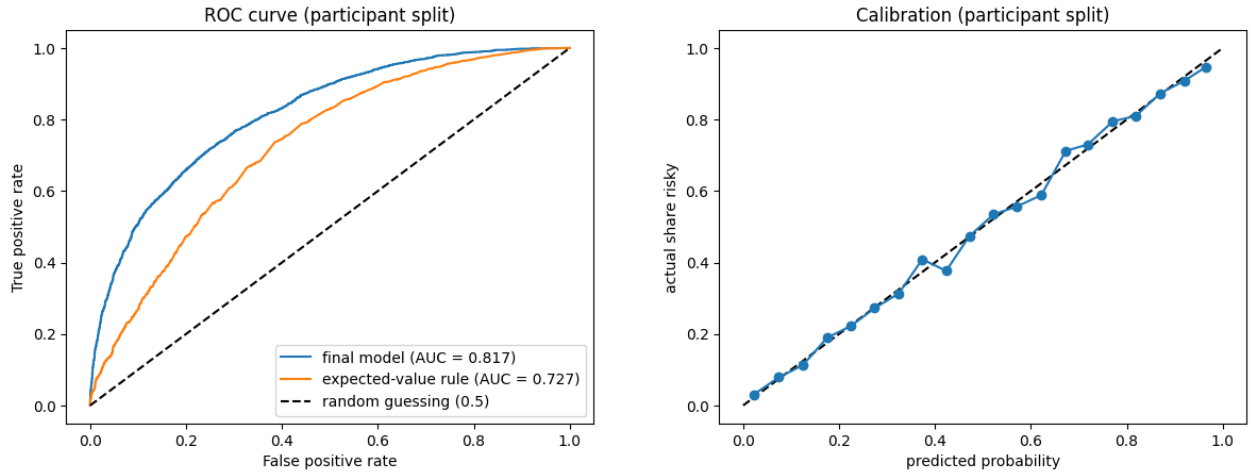


Figure 2: Left: ROC curve of the final model (participant split) against the EV rule. Right: reliability diagram of out-of-fold probabilities.

- Linear models only; I tried no tree-based or nonlinear models in this report.
- Associations only, not causal effects.

7 Conclusion

For unseen participants, framing and domain are the key information beyond the raw numbers (AUC 0.721 → 0.812), because an EV rule cannot see the frame. Prospect-theory features with a separate slope per domain/frame cell add a small further gain (0.817) that is within fold noise. A random split is not optimistic for a model without person identifiers, but becomes so (0.840 vs. 0.817) when identifiers are available, so grouped validation is the right choice for this task. Response time and person traits add nothing for new people.

References

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